

SPECIAL NOTE: To receive full credit, you must show all of your work and write your final answers in the spaces provided in this booklet.

Standards of Conduct- Academic Integrity

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Cheating on Examinations: Either intentionally using or attempting to use unauthorized materials, information, people, artificial intelligence, or study aids; providing to, or receiving from another person, any kind of unauthorized assistance on any examination.

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The above behaviors will not be tolerated and will result at a minimum in a zero on the exam or assignment. Other possible actions include an "F" on the final grade, investigation, and disciplinary action by the Office of the Dean of Students and in egregious cases, expulsion from UIC.

$$\begin{array}{r|l} 1 & 28 \\ \hline 2 & 49 \\ \hline T & 77 \end{array}$$

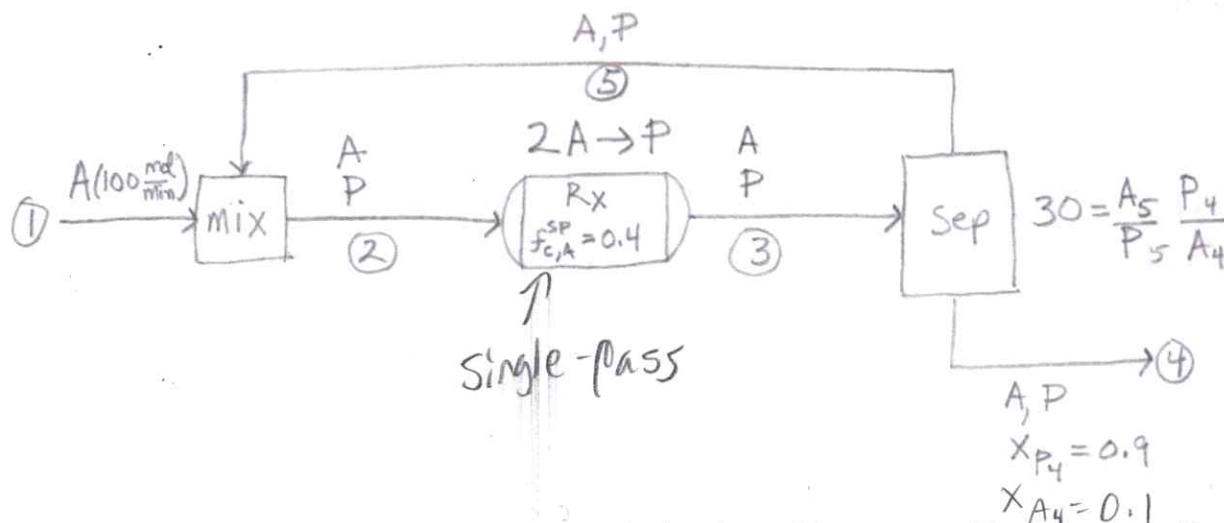
- (1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:



The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5 P_4}{P_5 A_4}$$

where the separation factor $\alpha_{AB} = 30$. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of $A_1 = 100 \text{ mol A/min}$.



- (a) Use a degree of freedom analysis to determine whether the problem as stated is well posed. (5 pts.)

4a) $1 + 2 + 2 + 2 + 2$

4b) $1 = 10$

5a) 1

5b) 1

5c) 2

5d) $2 + 2 + 2 = 10$

Well posed

$10 - 10 = 0$

DoF = 0

(Problem 1 Continued)

(b) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (40 pts.)

~~$$\dot{n}_{A, \text{out}} = \dot{n}_{A, \text{in}} = 25 \text{ mol/min}$$~~
~~$$\dot{n}_{P, \text{out}} = \dot{n}_{P, \text{in}} + \dot{n}_{A, \text{in}} = 25 + 25 = 50 \text{ mol/min}$$~~

$$f_{C, A}^{\text{SP}} = \frac{A_2 - A_3}{A_2} = 0.40 \quad \text{rct r} \rightarrow A_2 - 0.4A_2 = A_3 \quad A_3 = 0.6A_2$$

~~$$K_{AB} = \frac{A_5}{P_5} \cdot \frac{P_4}{A_4} = 30 = \frac{A_5}{P_5} \cdot \frac{0.9}{0.1}$$~~

Mixer: $A_2 = 100 + A_5$ ✓
 $P_2 = P_5$ ✓

missing Σ

rct r: $A_3 = 0.6A_2$ ✓
 $P_3 = P_2 + 0.8A_2$ ✓
 $A_3 = 60 + 0.6A_5$
 $P_3 = P_5 + 80 + 0.8A_5$

sep: $P_5 + P_4 = P_3$ ✓
 $A_5 + A_4 = A_3$ ✓

$$K = \frac{A_5}{P_5} \cdot \frac{P_4}{A_4} = 30$$

$$P_5 + P_4 = P_5 + 80 + 0.8A_5 \rightarrow P_4 = 80 + 0.8A_5$$

$$A_5 + A_4 = 60 + 0.6A_5 \rightarrow A_4 = 60 - 0.4A_5$$

$$\begin{aligned} 1.25P_4 &= 100 + A_5 \\ -2.5A_4 &= -150 + A_5 \\ \hline 1.25P_4 - 100 &= A_5 \\ -2.5A_4 + 150 &= A_5 \end{aligned}$$

$$\frac{A_5}{P_5} \cdot \frac{10}{1.11}$$

$$1.25P_4 + 2.5A_4 - 250 = 0$$

$$1.25P_4 + 2.5A_4 = 250$$

$$P_4 + 2A_4 = 200$$

$$K_{P_4} = 0.9 \rightarrow \frac{2A_4}{18} = P_4$$

$$K_{A_4} = 0.1$$

$$10 P_4 = 200$$

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$$P_4 = 10, A_4 = 1.11$$

(Problem 1 Continued)



Species↓/Steam→	1	2	3	4	5
A	100	100 + A₅	0.6 A₂	1.111	X
P	----	P ₅	P ₅ + 0.8 A ₂	1.0	X

(Problem 1 Continued)

(c) Calculate the overall fractional conversion based on species A (5 pts.)

$$f_{c,A}^{ov} = \frac{100 - A_4}{100}$$

$$f_{c,A}^{ov} = \underline{0.989}$$

3

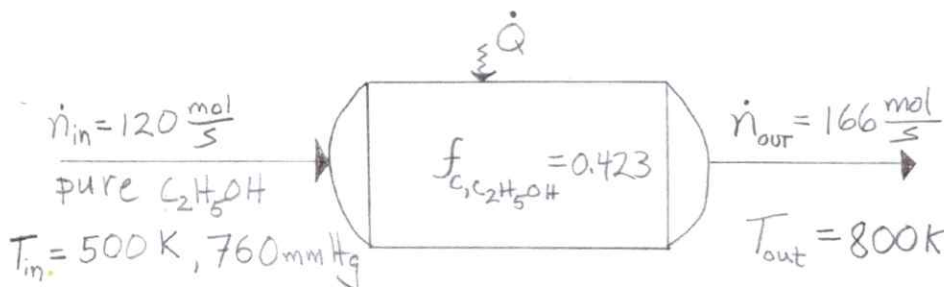
(2) (50 pts.) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$)



This process is carried out in a single reactor operating at steady state. The reactor has a feed rate of 120 mol/s of pure ethanol at 500K and 760 mm Hg. The reactor total product flow rate is 166 mol/s at 800K. The reactor converts 42.3% of the ethanol.



Data: The heat capacity in J/mol°C is given by $C_p = A + Bt$ where $t = T(\text{K})/1000$. The table below gives the constants A and B .

Species	A	B
$\text{C}_2\text{H}_5\text{OH}$	32.3	120
CH_3CHO	30.7	88.2
$\text{CH}_3\text{COOC}_2\text{H}_5$	60.9	191
H_2	28.4	1.54

- (a) Calculate the all the species molar flow rates for the product stream in mol/s and write them into the Input/Output Table (see next page). Also, calculate the two extents of reaction, ξ_1 and ξ_2 . (20 pts.)

$$\dot{n}_{\text{E out}} = \dot{n}_{\text{E in}} - \xi_1 - 2\xi_2 \rightarrow -50.76 = \xi_1 - 2\xi_2$$

$$\dot{n}_{\text{A out}} = \xi_1$$

$$\dot{n}_{\text{H out}} = \xi_1 + 2\xi_2$$

$$\dot{n}_{\text{EA out}} = \xi_2$$

$$\frac{\dot{n}_{\text{E in}} - \dot{n}_{\text{E out}}}{\dot{n}_{\text{E in}}} = 0.423$$

$$\dot{n}_{\text{E in}} - \dot{n}_{\text{E out}} = 0.423 \dot{n}_{\text{E in}}$$

$$166 = 120 + \xi_1 + \xi_2$$

$$46 = \xi_1 + \xi_2$$

$$50.76 = \xi_1 + 2\xi_2$$

$$\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 50.76 \\ 46 \end{bmatrix} \quad \text{ref}$$

$$\xi_1 = 41.24$$

$$\xi_2 = 4.76$$

$$\xi_1 = 41.24 \text{ mol/s} \quad \xi_2 = 4.76 \text{ mol/s}$$

$$\dot{n}_{\text{E in}} - 0.423 \dot{n}_{\text{E in}} = \dot{n}_{\text{E out}}$$

$$\dot{n}_{\text{E in}} = 120, \quad \dot{n}_{\text{E out}} = 69.24 \text{ mol/s}$$

20

(Problem 2 Continued)

- (b) Finish filling enthalpy values into the Input/Output Table below. Use a reference state of $T_{ref} = 25^\circ\text{C}$. (10 pts.)

Table B.3 Murphy

$$\hat{H}_1 = -234.95 \Delta \hat{H}_f^\circ + \frac{1}{1000} \int_{298}^{500} (32.3 + \frac{T}{1000} 120) dT = -218.8 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{H}_2 = \quad \quad \quad + \frac{1}{1000} \int_{298}^{800} 32.3 + \frac{T}{1000} 120 dT = -185.7 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{H}_3 = -166.2 \Delta \hat{H}_f^\circ + \frac{1}{1000} \int_{298}^{800} 30.7 + \frac{T}{1000} 88.2 dT = -126.5 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{H}_4 = -444.5 \Delta \hat{H}_f^\circ + \frac{1}{1000} \int_{298}^{800} 60.9 + \frac{T}{1000} 191 dT = -361.3 \frac{\text{kJ}}{\text{mol}}$$

$$\hat{H}_5 = 0 \Delta \hat{H}_f^\circ + \frac{1}{1000} \int_{298}^{800} 28.4 + \frac{T}{1000} 1.54 dT = 14.68 \frac{\text{kJ}}{\text{mol}}$$

10

Species	$\dot{n}_{in}$ (mol/s)	$\hat{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\hat{H}_{out}$ (kJ/mol)
C ₂ H ₅ OH	120	$\hat{H}_1 = -218.8$	69.24	$\hat{H}_2 = -185.7$
CH ₃ CHO	----	----	41.24	$\hat{H}_3 = -126.5$
CH ₃ COOC ₂ H ₅	----	----	50.76	$\hat{H}_4 = -361.3$
H ₂	----	----	4.76	$\hat{H}_5 = 14.68 \frac{\text{kJ}}{\text{mol}}$

120

166

(Problem 2 Continued)

- (c) Calculate the heat rate in kJ/s for the reactor. Indicate clearly whether it is heat in or out of the reactor. (20 pts.)

$$\dot{Q} = \Delta \hat{H} = \sum_i \dot{n}_{out,i} \hat{H}_{out,i} - \sum_i \dot{n}_{in,i} \hat{H}_{in,i}$$

$$(-185.7 \cdot 69.24) + (-126.5 \cdot 41.24) + (-361.3 \cdot 50.76) + (14.68 \cdot 4.76) - (-218.8 \cdot 120) = -10088.4 \frac{\text{kJ}}{\text{s}} \text{ from reactor}$$

you must add heat to it

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~~scribble~~

$$\dot{Q} = \frac{10088.4}{\text{To reactor}} \text{ kJ/s}$$

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